Product Requirements Document

C-ITS PKI Implementation

Certificate Generation conforming to ETSI TS 103 097 V2.2.1

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1. Executive Summary

This Product Requirements Document (PRD) specifies the functional and technical requirements for implementing a simple Public Key Infrastructure (PKI) capable of generating and managing digital certificates for Cooperative Intelligent Transport Systems (C-ITS). The implementation shall conform to ETSI TS 103 097 V2.2.1 — "Intelligent Transport Systems (ITS); Security; Security header and certificate formats; Release 2" — which itself is defined as a profile of IEEE Std 1609.2™-2025.

The PKI system must support the full certificate hierarchy required by the C-ITS trust model: a self-signed Root Certification Authority (Root CA), subordinate Enrolment Authority (EA) and Authorization Authority (AA) certificates, as well as end-entity Enrolment Credentials (EC) and Authorization Tickets (AT) issued to ITS-Stations. A Trust List Manager (TLM) certificate is also required. All certificates shall use the EtsiTs103097Certificate format encoded with COER (Canonical Octet Encoding Rules) as mandated by the specification.

2. Background and Context

Cooperative Intelligent Transport Systems rely on vehicles, roadside units, and infrastructure nodes exchanging safety-critical messages such as Cooperative Awareness Messages (CAMs) and Decentralized Environmental Notification Messages (DENMs). These messages must be digitally signed to ensure authenticity and integrity, and optionally encrypted for confidentiality. The trust framework enabling this relies on a multi-level PKI that issues short-lived pseudonym certificates (Authorization Tickets) to ITS-Stations so that vehicle identity is protected while message authenticity is guaranteed.

ETSI TS 103 097 V2.2.1 defines the precise data structures, certificate formats, cryptographic algorithms, and security profiles that all conformant C-ITS implementations must use. It builds upon IEEE Std 1609.2 as the base data format standard and adds ETSI-specific profiling constraints.

2.1 Normative References

Ref.	Document
[1]	IEEE Std 1609.2™-2025 — Wireless Access in Vehicular Environments – Security Services for Application and Management Messages
[2]	ETSI TS 102 965 — ITS Application Object Identifier (ITS-AID); Registration; Release 2
[3]	ITU-T Recommendation X.696 — ASN.1 Canonical Octet Encoding Rules (COER/OER)

2.2 Informative References

Ref.	Document
[i.1]	ETSI TS 102 940 — ITS Communications Security Architecture and Security Management;

	Release 2
[i.2]	ETSI TS 102 941 — ITS Security; Trust and Privacy Management; Release 2
[i.3]	ETSI TS 103 601 — ITS Security; Security Management Messages Communication Requirements and Distribution Protocols; Release 2
[i.4]	CPOC Protocol
[i.5]	ETSI TS 103 096-1 — ITS Testing; Conformance Test Specifications for ITS Security Part 1 (PICS); Release 2

3. Scope

3.1 In Scope

The following capabilities are within the scope of this implementation:

- Root CA certificate generation (self-signed, explicit certificate type)
- Enrolment Authority (EA) subordinate CA certificate generation
- Authorization Authority (AA) subordinate CA certificate generation
- Trust List Manager (TLM) certificate generation
- Enrolment Credential (EC) issuance for ITS-Stations
- Authorization Ticket (AT) issuance for ITS-Stations
- EtsiTs103097Certificate format conformance (clause 6 and 7.2 of ETSI TS 103 097)
- ECDSA signing with NIST P-256 and P-384 curves
- AES-128-CCM symmetric encryption and ECIES key wrapping
- COER (Canonical OER) encoding of all ASN.1 data structures
- Signed data structure generation conforming to EtsiTs103097Data-Signed profile
- Encrypted data structure generation conforming to EtsiTs103097Data-Encrypted profile
- Certificate validity period management
- Permissions management (appPermissions and certIssuePermissions)
- Hash ID-based revocation support for EA and AA certificates

3.2 Out of Scope

The following are explicitly out of scope for this implementation:

- Online Certificate Status Protocol (OCSP)
- Full CPOC (Central Point of Contact) protocol implementation
- Certificate Revocation List (CRL) distribution infrastructure (format defined in ETSI TS 102 941)
- Certificate Trust List (CTL) distribution infrastructure
- Misbehaviour Detection Management (MDM) processing logic
- V2X radio communication stack
- ITS-S on-board application integration

- Web-based management portal or GUI

4. Glossary and Abbreviations

Abbreviation / Term	Definition
AA	Authorization Authority — subordinate CA that issues Authorization Tickets
ASN.1	Abstract Syntax Notation One — formal notation for data structure schemas
AT	Authorization Ticket — short-lived pseudonym certificate issued to an ITS-Station
CAM	Cooperative Awareness Message — periodic V2X safety broadcast
C-ITS	Cooperative Intelligent Transport Systems
COER	Canonical Octet Encoding Rules — the required ASN.1 encoding per ITU-T X.696
CPOC	Central Point of Contact — operator coordinating the ETSI C-ITS PKI
CTL	Certificate Trust List — signed list of trusted Root CA certificates
CRL	Certificate Revocation List
DENM	Decentralized Environmental Notification Message — event-driven V2X message
EA	Enrolment Authority — subordinate CA that issues Enrolment Credentials
EC	Enrolment Credential — long-term identity certificate for an ITS-Station
ECDSA	Elliptic Curve Digital Signature Algorithm
ECIES	Elliptic Curve Integrated Encryption Scheme
ITS	Intelligent Transport Systems
ITS-AID	ITS Application Identifier — registered per ETSI TS 102 965
ITS-S	ITS-Station (vehicle OBU, RSU, or back-end node)
KDF	Key Derivation Function
MDM	Misbehaviour Detection Management
OBU	On-Board Unit
OID	Object Identifier
PICS	Protocol Implementation Conformance Statement
PKI	Public Key Infrastructure
Root CA	Root Certification Authority — the trust anchor of the hierarchy
RSU	Road-Side Unit
TLM	Trust List Manager — entity managing the Certificate Trust List
V2X	Vehicle-to-Everything communications

5. System Architecture Overview

The C-ITS PKI follows a three-tier hierarchy. The Root CA is the trust anchor and signs subordinate CA certificates. Two categories of subordinate CA exist: the Enrolment Authority (EA), which issues long-lived Enrolment Credentials to ITS-Stations, and the Authorization Authority (AA), which issues short-lived, pseudonymous Authorization Tickets. A Trust List Manager (TLM) certificate establishes the signed Certificate Trust List. Each certificate type must conform to a specific security profile defined in clause 7.2 of ETSI TS 103 097 V2.2.1.

5.1 PKI Hierarchy

Entity	Certificate Type	Issuer	Issues
Trust List Manager (TLM)	TLM Certificate	Self (issuer = self)	Signs CTLs
Root CA	Root CA Certificate	Self (issuer = self)	EA Cert, AA Cert
Enrolment Authority (EA)	Sub-CA Certificate	Root CA	Enrolment Credentials
Authorization Authority (AA)	Sub-CA Certificate	Root CA	Authorization Tickets
ITS-Station (enrolment)	Enrolment Credential (EC)	EA	N/A — end entity
ITS-Station (pseudonym)	Authorization Ticket (AT)	AA	N/A — end entity

5.2 Data Format Foundation

All data structures shall be defined using ASN.1 as specified in the normative ASN.1 module EtsiTs103097Module. The module imports from the IEEE Std 1609.2 modules Ieee1609Dot2 and Ieee1609Dot2BaseTypes. All encoding shall use COER as required by ITU-T X.696. The module OID is:

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{itu-t(0) identified-organization(4) etsi(0) itsDomain(5) wg5(5)
 secHeaders(103097) core(1) major-version-3(3) minor-version-2(2)}
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6. Cryptographic Algorithm Requirements

The following cryptographic algorithms are mandated by ETSI TS 103 097 V2.2.1 clause 4.2. All implementations shall support these algorithms; no others shall be used for interoperability-critical operations.

Algorithm Category	Algorithm	Standard Reference	Notes
Digital Signature	ECDSA	IEEE Std 1609.2 §5.3.1	P-256 (SHA-256) or P-384 (SHA-384)
Hash (signature)	SHA-256	IEEE Std 1609.2	Used with P-256 curve
Hash (signature)	SHA-384	IEEE Std 1609.2	Used with P-384 curve

Public Key Encryption	ECIES	IEEE Std 1609.2 §5.3.5	Elliptic Curve Integrated Encryption Scheme
Symmetric Encryption	AES-128-CCM	IEEE Std 1609.2 §5.3.8	16-octet key; 12-octet nonce; CCM mode
Key Derivation	KDF2 based on SHA-256	IEEE Std 1609.2	TRUNCATE(SHA256(S||0x00000001||P1) || SHA256(S||0x00000002||P1), 48)
MAC	HMAC based on KDF2 output	FIPS PUB 198-1	km = last 32 bytes of KDF2; MAC(m,km) = HMAC(m,km)

6.1 ECIES Key Encapsulation Details

The ECIES scheme used for key wrapping shall implement the following parameters:

- Elliptic curve: NIST P-256 (or P-384 for SHA-384 profiles); base point G, order q
- Sender generates ephemeral private key r ∈ [1, q-1]; ephemeral public key V = r·G (33 octets, compressed)
- Shared secret: S = Px, where (Px, Py) = r·Kr and Kr is the receiver's public encryption key
- KDF output: (ke ‖ km) = KDF2(S), where ke is 16 octets and km is 32 octets
- AES key encryption: c = E(A, ke) using XOR — c = A ⊕ ke (16 octets)
- Authentication tag: t = MAC(c, km) = HMAC(c, km), 16 octets

6.2 AES-CCM Message Encryption

- Key A: 16 octets, randomly generated per message
- Nonce n: 12 octets, randomly generated
- AES-CCM encrypts message m producing ciphertext C = encrypted-m ‖ CCM-authentication-tag
- Transmitted alongside: V (ephemeral public key), c (encrypted AES key), t (ECIES auth tag), nonce n, cert_id (8 octets)

7. Secure Data Structure Requirements

A secure data structure shall be of type EtsiTs103097Data, corresponding to Ieee1609Dot2Data as defined in IEEE Std 1609.2, clause 6.3.2, with the constraints defined in clauses 5.1–5.3 of ETSI TS 103 097 V2.2.1.

7.1 Supported Data Structure Variants

Type Name	Description	Clause
EtsiTs103097Data-Unsecured	Wraps unsecured (plain) data payload	5.1
EtsiTs103097Data-Signed	Signed message; payload in tbsData.payload.data	5.1, 5.2
EtsiTs103097Data-	Signed message with external payload; digest in	5.1, 5.2

SignedExternalPayload	tbsData.payload.extDataHash	
EtsiTs103097Data-Encrypted	AES-CCM encrypted payload in ciphertext.aes128ccm	5.1, 5.3
EtsiTs103097Data-SignedAndEncrypted	Signed message wrapped inside an Encrypted structure	5.1
EtsiTs103097Data-Encrypted-Unicast	Encrypted with exactly one entry in recipients	5.1
EtsiTs103097Data-SignedAndEncrypted-Unicast	Signed and encrypted with exactly one recipient	5.1

7.2 SignedData Constraints

The type SignedData shall conform to the following constraints (clause 5.2):

- hashId: indicates the hash algorithm (sha256 or sha384) per IEEE Std 1609.2 clauses 6.3.5 and 5.3.3
- tbsData shall be of type ToBeSignedData (IEEE Std 1609.2 clause 6.3.6)
- tbsData.payload shall be of type SignedDataPayload containing either data (Ieee1609Dot2Data) or extDataHash
- tbsData.headerInfo.psid: shall contain the ITS-AID for the relevant message type (registered in ETSI TS 102 965)
- tbsData.headerInfo.generationTime: always present
- tbsData.headerInfo.expiryTime: present or absent per message security profile (clause 7.1)
- tbsData.headerInfo.generationLocation: present or absent per message security profile
- tbsData.headerInfo.p2pcdLearningRequest: always absent
- tbsData.headerInfo.missingCrlIdentifier: always absent
- tbsData.headerInfo.encryptionKey: present or absent per message security profile
- signer shall be of type SignerIdentifier, constrained to: (a) digest — 8-octet hash of the signing AT/EC certificate, or (b) certificate — a single EtsiTs103097Certificate
- signature shall contain the ECDSA signature per IEEE Std 1609.2 clauses 6.3.38, 6.3.39 and 5.3.1

7.3 EncryptedData Constraints

The type EncryptedData shall conform to the following constraints (clause 5.3):

- recipients shall be of type SequenceOfRecipientInfo; each entry shall be pskRecipInfo, certRecipInfo, or signedDataRecipInfo
- Encryption scheme: ECIES as defined in IEEE Std 1609.2 clause 5.3.5
- ciphertext shall be of type SymmetricCiphertext containing an EtsiTs103097Data encrypted per AES-CCM
- The inner encrypted payload shall be an EtsiTs103097Data structure

8. Certificate Format Requirements

All certificates shall be of type EtsiTs103097Certificate corresponding to a single Certificate as defined in IEEE Std 1609.2 clause 6.4.2, with the constraints defined in clause 6 of ETSI TS 103 097 V2.2.1. A certificate shall be either explicit (verifyKeyIndicator = verificationKey) or implicit (verifyKeyIndicator = reconstructionValue) at the discretion of the issuing entity. Receivers shall be able to validate both types.

8.1 toBeSigned Component Constraints

The component toBeSigned (of type ToBeSignedCertificate) shall be constrained as follows:

- id: of type CertificateId, constrained to choice name (for CA and EC certificates) or none (for AT certificates)
- cracaId: set to 000000H (revocation via Hash ID method defined in ETSI TS 102 941)
- crlSeries: set to 0D
- validityPeriod: the validity interval [start, start + duration) — start inclusive, end exclusive; no further constraints
- region: of type GeographicRegion, present or absent per certificate profile; the value 65535 designates the European Union (27 member states as of 31 January 2020)
- assuranceLevel: of type SubjectAssurance, present or absent per certificate profile
- appPermissions: of type SequenceOfPsidSsp, present or absent per certificate profile; at least one of appPermissions or certIssuePermissions shall be present
- certIssuePermissions: of type SequenceOfPsidGroupPermissions, present or absent per certificate profile
- certRequestPermissions: always absent
- canRequestRollover: always absent
- encryptionKey: of type PublicEncryptionKey, present or absent per certificate profile
- verifyKeyIndicator: verificationKey if explicit certificate; reconstructionValue if implicit
- flags: present or absent per [i.2]
- appExtensions: present only if AppExtension OperatingOrganizationId identified by certExtId-OperatingOrganization is present
- certIssueExtensions: absent, or optionally present only if appExtensions is present
- certRequestExtensions: absent, or optionally present only if certIssueExtensions is present

8.2 Signature Component

The component signature of EtsiTs103097Certificate shall be of type Signature as defined in IEEE Std 1609.2 clause 6.3.37, and shall contain the ECDSA signature calculated by the signer identified in the issuer component (clauses 6.3.38, 6.3.39, and 5.3.1 of IEEE Std 1609.2).

8.3 Certificate Validity Assessment

The validity of a certificate shall be assessed as defined in IEEE Std 1609.2 clause 5.1, using the Hash ID-based revocation method for EA and AA certificates. No revocation method applies

to Authorization Tickets and Enrolment Credentials. Root CA certificate validity against the CTL is governed by ETSI TS 102 941.

9. Certificate Profile Requirements

Each certificate type shall implement a specific security profile as defined in clause 7.2 of ETSI TS 103 097 V2.2.1. The following sub-sections specify the mandatory constraints for each profile.

9.1 Root CA Certificate

- Certificate type: explicit (verifyKeyIndicator = verificationKey)
- issuer: set to self
- CertificateId (toBeSigned.id): choice name — shall contain a unique name string for the root CA
- certIssuePermissions: present — indicates issuing permissions (permissions to sign subordinate CA certificates)
- appPermissions: present — shall include ITS-AID for the CRL service and ITS-AID for the CTL service (registered per ETSI TS 102 965)
- encryptionKey: not required (absent)
- Additional requirements: per CPOC protocol [i.4]

9.2 Enrolment Authority (EA) Certificate

- Certificate type: explicit
- issuer: sha256AndDigest or sha384AndDigest — digest of the signing Root CA certificate
- CertificateId: choice name — unique name for the EA
- certIssuePermissions: present — indicates permissions to sign Enrolment Credentials
- appPermissions: present — indicates permissions to sign certificate response messages (EtsiTs103097Data)
- encryptionKey: present — public encryption key for ITS-Stations to encrypt EC request messages to the EA
- Additional requirements: per CPOC protocol [i.4]

9.3 Authorization Authority (AA) Certificate

- Certificate type: explicit
- issuer: sha256AndDigest or sha384AndDigest — digest of the signing Root CA certificate
- CertificateId: choice name — unique name for the AA, or choice none
- certIssuePermissions: present — indicates permissions to sign Authorization Tickets
- appPermissions: present — indicates permissions to sign certificate response messages (EtsiTs103097Data)
- encryptionKey: present — public encryption key for ITS-Stations to encrypt AT request messages to the AA

- Additional requirements: per CPOC protocol [i.4]

9.4 Trust List Manager (TLM) Certificate

- Certificate type: explicit
- issuer: set to self
- CertificateId: choice name — unique name string for the TLM
- appPermissions: present — shall contain the ITS-AID for the CTL service (per ETSI TS 102 965)
- encryptionKey: absent
- certIssuePermissions: absent
- Additional requirements: per CPOC protocol [i.4]

9.5 Enrolment Credential (EC)

- Certificate type: explicit
- issuer: sha256AndDigest or sha384AndDigest — digest of the signing EA certificate
- CertificateId: choice name — unique name associated with the enrolment credential
- appPermissions: present — indicates permissions to sign certificate request messages (EtsiTs103097Data)
- certIssuePermissions: absent
- The EC is the long-term identity credential for the ITS-Station; it is used to obtain ATs

9.6 Authorization Ticket (AT)

- Certificate type: explicit or implicit — at the discretion of the AA
- issuer: sha256AndDigest or sha384AndDigest — digest of the signing AA certificate
- CertificateId: choice none (pseudonymous — no name)
- appPermissions: present — indicates message signing permissions (permissions to sign an EtsiTs103097Data)
- certIssuePermissions: absent
- ATs are short-lived pseudonym certificates used for V2X message signing; no linkage to vehicle identity shall be inferable

9.7 Misbehaviour Authority Certificate

- Certificate type: explicit
- issuer: sha256AndDigest or sha384AndDigest
- CertificateId: choice name — unique name for the misbehaviour authority
- encryptionKey: present — public key for ITS-Stations to encrypt misbehaviour reports
- certIssuePermissions: absent
- appPermissions: present — shall include at least the ITS-AID for the Misbehaviour Detection Management (MDM) service

10. Message Security Profile Requirements

Clause 7.1 of ETSI TS 103 097 V2.2.1 defines security profiles for messages processed by the PKI. The PKI implementation shall generate and validate signed data structures conforming to the following profiles.

10.1 CAM Security Profile

Secure data structures containing Cooperative Awareness Messages shall conform to EtsiTs103097Data-Signed with:

- signer: by default choice digest (8-octet hash of the AT); choice certificate (full AT) included once per second after last inclusion, and immediately upon receiving a CAM signed by an unknown AT
- headerInfo.psid: ITS-AID value for CAMs as assigned in ETSI TS 102 965
- headerInfo.generationTime: always present
- headerInfo.inlineP2pcdRequest: shall be included when the ITS-S receives a digest pointing to an unknown AT, or receives a message with a certificate signed by an unknown AA certificate
- headerInfo.requestedCertificate: included if the ITS-S finds a digest of a valid AA certificate in an inlineP2pcdRequest received from a peer
- All other headerInfo components allowed by clause 5: absent

10.2 DENM Security Profile

Secure data structures containing Decentralized Environmental Notification Messages shall conform to EtsiTs103097Data-Signed with:

- signer: choice certificate — shall contain a full EtsiTs103097Certificate
- headerInfo.psid: ITS-AID value for DENMs as assigned in ETSI TS 102 965
- headerInfo.generationTime: always present
- headerInfo.generationLocation: shall be present
- All other headerInfo components: absent

10.3 Generic Signed Message Profile

Signed messages other than CAM and DENM shall use either EtsiTs103097Data-Signed (full payload) or EtsiTs103097Data-SignedExternalPayload (digest only), with constraints from clause 5.2.

10.4 Encrypted Message Profile

Encrypted messages shall use type EtsiTs103097Data-Encrypted with constraints from clause 5.3.

10.5 Signed and Encrypted Message Profile

Messages that are both signed and encrypted shall use EtsiTs103097Data-SignedAndEncrypted. This is an EtsiTs103097Data-Encrypted containing an EtsiTs103097Data-Signed as the plaintext payload.

11. Functional Requirements

11.1 Key Pair Generation

ID	Requirement
FR-KG-01	The system shall generate ECDSA P-256 key pairs for use with SHA-256
FR-KG-02	The system shall generate ECDSA P-384 key pairs for use with SHA-384
FR-KG-03	Private keys shall be generated using a cryptographically secure random number generator
FR-KG-04	Public keys shall be stored in compressed point format (33 octets for P-256, 49 for P-384)
FR-KG-05	Separate key pairs shall be generated for signing and encryption for each PKI entity

11.2 Certificate Issuance

ID	Requirement
FR-CI-01	The system shall generate self-signed Root CA certificates conforming to profile 9.1
FR-CI-02	The system shall generate EA certificates signed by the Root CA conforming to profile 9.2
FR-CI-03	The system shall generate AA certificates signed by the Root CA conforming to profile 9.3
FR-CI-04	The system shall generate TLM certificates (self-signed) conforming to profile 9.4
FR-CI-05	The system shall issue Enrolment Credentials signed by the EA conforming to profile 9.5
FR-CI-06	The system shall issue Authorization Tickets signed by the AA conforming to profile 9.6
FR-CI-07	All certificates shall be of type EtsiTs103097Certificate encoded in COER
FR-CI-08	Certificate validity periods shall be configurable; the interval is [start, start+duration)
FR-CI-09	At least one of appPermissions or certIssuePermissions shall be present in each certificate
FR-CI-10	cracaId shall be set to 000000H and crlSeries to 0D in all certificates
FR-CI-11	The system shall support both explicit and implicit certificate types and receivers shall validate both

11.3 Signing Operations

ID	Requirement
FR-SN-01	The system shall produce SignedData structures of type EtsiTs103097Data-Signed
FR-SN-02	Signatures shall use ECDSA per IEEE Std 1609.2 clause 5.3.1
FR-SN-03	hashId shall match the curve: sha256 with P-256, sha384 with P-384
FR-SN-04	The signer field shall be either digest (8-octet cert hash) or certificate (full EtsiTs103097Certificate)
FR-SN-05	headerInfo.generationTime shall always be present in signed data
FR-SN-06	p2pcdLearningRequest and missingCrlIdentifier shall always be absent
FR-SN-07	The system shall support external payload signing (EtsiTs103097Data-SignedExternalPayload)

11.4 Encryption Operations

ID	Requirement
FR-EN-01	The system shall produce EncryptedData structures using ECIES for key encapsulation
FR-EN-02	Message payload shall be encrypted with AES-128-CCM (16-octet key, 12-octet nonce)
FR-EN-03	The system shall support pskRecipInfo, certRecipInfo, and signedDataRecipInfo recipient types
FR-EN-04	The system shall support unicast encryption (exactly one recipient)
FR-EN-05	The ephemeral ECDSA key V shall be freshly generated per encryption operation
FR-EN-06	The system shall implement the ECIES decryption procedure as specified in Annex B.3

11.5 Certificate Verification

ID	Requirement
FR-VF-01	The system shall verify certificate signatures using ECDSA
FR-VF-02	Certificate validity shall be assessed per IEEE Std 1609.2 clause 5.1
FR-VF-03	The system shall verify SignedData signatures and reconstruct the message hash
FR-VF-04	Revocation checking for EA and AA certificates shall use the Hash ID-based method
FR-VF-05	The system shall correctly parse both explicit and implicit certificates
FR-VF-06	The system shall implement region constraint checking; value 65535 = EU-27

11.6 Permissions Management

ID	Requirement
FR-PM-01	appPermissions shall encode ITS-AID values using PSID as defined in ETSI TS 102 965
FR-PM-02	certIssuePermissions shall encode PsidGroupPermissions constraining what sub-certificate permissions may be granted
FR-PM-03	An AT shall carry appPermissions only (no certIssuePermissions)
FR-PM-04	An EC shall carry appPermissions for certificate request message signing
FR-PM-05	Root CA shall carry certIssuePermissions for signing subordinate CAs and appPermissions for CRL/CTL service AIDs
FR-PM-06	EA and AA shall carry certIssuePermissions for signing end-entity certificates and appPermissions for certificate response message signing

12. Non-Functional Requirements

12.1 Security

ID	Requirement
NFR-SEC-01	Private keys shall never be stored in plaintext outside of a secure element or HSM
NFR-SEC-02	All random number generation shall use a CSPRNG meeting at least NIST SP 800-90A Rev. 1 requirements
NFR-SEC-03	The system shall prevent timing side-channel attacks on ECDSA and ECIES operations
NFR-SEC-04	AES-CCM nonces shall be unique per encryption operation; nonce reuse with the same key is prohibited
NFR-SEC-05	Authorization Ticket private keys shall be independent of Enrolment Credential private keys
NFR-SEC-06	The system shall not expose vehicle identity through AT certificate fields; AT id shall be choice none

12.2 Interoperability

ID	Requirement
NFR-INT-01	All ASN.1 data structures shall be encoded using COER (ITU-T X.696)
NFR-INT-02	The implementation shall conform to ETSI TS 103 097 V2.2.1 PICS proforma (ETSI TS 103 096-1)
NFR-INT-03	The implementation shall correctly handle UnCountryId, CountryAndRegions, CountryAndSubregions, and RegionAndSubregions by parsing (not necessarily recognizing all values)

NFR-INT-04	OID of the ETSI ASN.1 module shall match the value specified in Annex A.1 of the standard
NFR-INT-05	The system shall be backwards compatible with explicit certificates from prior versions (V1.3.1 and V1.4.1)

12.3 Performance

ID	Requirement
NFR-PER-01	AT issuance latency shall not exceed 500 ms per certificate under nominal conditions
NFR-PER-02	CAM signing latency shall not exceed 10 ms for ECDSA P-256 signing
NFR-PER-03	The system shall support concurrent certificate issuance requests (minimum 10 concurrent)
NFR-PER-04	Certificate verification latency shall not exceed 50 ms per certificate chain

12.4 Reliability and Availability

ID	Requirement
NFR-REL-01	The PKI backend (Root CA, EA, AA) shall target 99.9% availability during operational hours
NFR-REL-02	Root CA key material shall be stored with hardware redundancy and offline backup
NFR-REL-03	Certificate issuance operations shall be atomic — partial issuance failures shall not produce malformed certificates

13. ASN.1 Encoding Requirements

All data structures shall be defined and encoded in accordance with the normative ASN.1 modules referenced in Annex A of ETSI TS 103 097 V2.2.1.

13.1 ASN.1 Module References

Module Name	Module OID	SHA-256 Hash
EtsiTs103097Module	{itu-t(0) id-org(4) etsi(0) itsDomain(5) wg5(5) secHeaders(103097) core(1) major-version-3(3) minor-version-2(2)}	245a3c10c176497f658c6a4d9ec804d3226dda2ac10a4351e382ddb5afe9ff72
EtsiTs103097Extension	{itu-t(0) id-org(4) etsi(0)	b94c9b373567dd9bfb15d61c8c206d5630f6b0c481ef41ddd5

Module	itsDomain(5) wg5(5) secHeaders(103097) extension(2) major-version-1(1) minor-version-2(2)}	74c96784a9eb1a
Ieee1609Dot2	{iso(1) id-org(3) ieee(111) standards-assoc-numbered-series-stds(2) wave-stds(1609) dot2(2) base(1) schema(1) major-version-2(2) minor-version-7(7)}	82b5e35cbaadae1c6b2f087626b10d52afc73779c9ac6700c6445830d8824e0b
Ieee1609Dot2BaseTypes	{iso(1) id-org(3) ieee(111) standards-assoc-numbered-series-stds(2) wave-stds(1609) dot2(2) base(1) base-types(2) major-version-2(2) minor-version-5(5)}	bf2b3d66d394449319323f8a59d8084d963c3ad55d6790e1436ab897221690fe

13.2 Encoding Rules

- All ASN.1 data structures shall be encoded using Canonical OER (COER) as defined in ITU-T X.696
- For structures with a "canonical encoding" in IEEE Std 1609.2, that canonical encoding shall be used
- The ASN.1 modules shall be obtained from the ETSI forge repository and integrity verified against SHA-256 hashes in Table 13.1
- Extension mechanisms (ASN.1 extensibility) shall be supported per clause 4.3 of ETSI TS 103 097 V2.2.1
- HeaderInfo extensions shall use the contributedExtensions component with etsiHeaderInfoContributorId = 2

14. Acceptance Criteria

The implementation shall be considered accepted when all of the following criteria are demonstrated through testing:

AC ID	Acceptance Criterion	Test Method
AC-01	Root CA certificate passes COER decode and conforms to profile 9.1	Automated ASN.1 decode + profile check
AC-02	EA certificate is verifiable against Root CA public key	ECDSA signature verification test
AC-03	AA certificate is verifiable against Root CA public key	ECDSA signature verification test
AC-04	EC issued by EA is verifiable against EA public key	ECDSA signature verification test
AC-05	AT issued by AA is verifiable against AA public key; id = none	ECDSA + field constraint test
AC-06	CAM signed with AT passes SignedData verification per CAM profile	End-to-end signing and verification
AC-07	DENM signed with AT includes generationLocation and signer=certificate	Field presence test
AC-08	AES-CCM encrypted message decrypts correctly with ECIES key recovery	Encrypt/decrypt round-trip test
AC-09	Implicit certificate reconstruction produces correct verification key	Reconstruction + ECDSA verify test
AC-10	Region constraint 65535 is accepted and interpreted as EU-27	Region parsing test
AC-11	cracaId = 000000H and crlSeries = 0D in all generated certificates	Field value inspection
AC-12	All structures decode without error using a conformant IEEE 1609.2 COER decoder	Cross-implementation interop test
AC-13	PICS proforma (ETSI TS 103 096-1) conformance claims are substantiated	PICS review against test results

Appendix A: Certificate Issuance Workflow Summary

A.1 PKI Initialization Sequence

1. Generate Root CA key pair (ECDSA P-256 or P-384)
2. Generate Root CA self-signed certificate (issuer = self; certIssuePermissions present; appPermissions with CRL and CTL ITS-AIDs)
3. Generate TLM key pair and TLM self-signed certificate (issuer = self; appPermissions with CTL ITS-AID)
4. Generate EA key pair (signing + encryption)

5. EA submits certificate request to Root CA; Root CA issues EA certificate (certIssuePermissions + encryptionKey present)
6. Generate AA key pair (signing + encryption)
7. AA submits certificate request to Root CA; Root CA issues AA certificate (certIssuePermissions + encryptionKey present)

A.2 ITS-Station Enrolment Sequence

8. ITS-S generates EC key pair
9. ITS-S encrypts enrolment request to EA (using EA encryptionKey; EtsiTs103097Data-Encrypted)
10. EA validates ITS-S identity and issues Enrolment Credential (EC) signed by EA
11. EC id = choice name; appPermissions for certificate request message signing; certIssuePermissions absent

A.3 Authorization Ticket Provisioning Sequence

12. ITS-S generates fresh AT key pair(s)
13. ITS-S encrypts AT request to AA (using AA encryptionKey; signed with EC inside EtsiTs103097Data-Encrypted)
14. AA validates EC and issues Authorization Ticket (AT) signed by AA
15. AT id = choice none; appPermissions for message signing; certIssuePermissions absent
16. ITS-S uses AT private key to sign CAMs and DENMs; includes AT digest (default) or full AT certificate in signer field